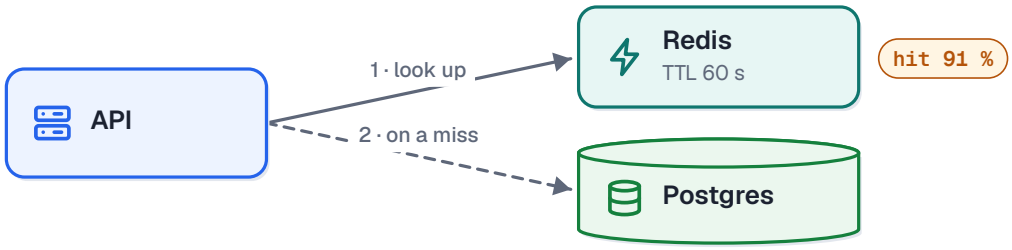
Caching order reads

A read of one order is answered from Redis when it can be, and from Postgres when it cannot. This handout says how, what it costs, and what to watch.

The path



1. The API looks the order up in Redis.
2. On a miss it reads Postgres, and writes the row back to Redis for 60 seconds.
3. The next read of the same order is a hit.

What it costs

Read	p50	p99
Hit	9 ms	13 ms
Miss	41 ms	48 ms

A miss is a slower read, not an error. What matters is the **hit rate**: each point lost moves about 6 000 reads a day onto Postgres.

The code

```
def get_order(order_id: int) -> Order:
    cached = redis.get(f"order:{order_id}")
    if cached is not None:                      # a hit: 13 ms at p99
        return Order.parse(cached)
    order = db.fetch_order(order_id)            # a miss: Postgres
    redis.set(f"order:{order_id}", order.dump(), ex=60)
    return order
```

Read more

The design note, with the sequence of a miss and the week's numbers, is one scan away: